

HYPERBOLIC GEOMETRY — EXERCISES FROM LECTURE 1

Exercise 1.A

Let γ be a path of the form

$$\gamma : t \mapsto x_0 + iy(t)$$

for some $x_0 \in \mathbb{R}$ and some strictly increasing piecewise C^1 function $y : [a, b] \rightarrow (0, +\infty)$. Compute $|\gamma|_{\text{hyp}}$.

Exercise 1.B

Let $\gamma : [a, b] \rightarrow \mathbb{H}^2$ be a piecewise C^1 path, and let $\phi : [a', b'] \rightarrow [a, b]$ be a strictly increasing C^1 function. Prove that $|\gamma \circ \phi|_{\text{hyp}} = |\gamma|_{\text{hyp}}$.

Exercise 1.C

What is the topology on $\mathbb{H}^2$ induced by d_{hyp} ?

Exercise 1.D

Prove that the following maps are isometries of $\mathbb{H}^2$:

- (i) Horizontal translations,
- (ii) Reflections along vertical lines,
- (iii) Homotheties.

Deduce that the action of the isometry group $\text{Isom}(\mathbb{H}^2) \curvearrowright \mathbb{H}^2$ is transitive.

Exercise 1.E

The hyperbolic line is the set $\mathbb{H}^1 := (0, +\infty)$, with local hyperbolic norm given by $\|v\|_{\text{hyp}}^y = \frac{|v|}{y}$ for $y \in \mathbb{H}^1$ and $v \in \mathbb{R}$. The hyperbolic length of a (piecewise) C^1 path and the hyperbolic distance are defined analogously to the 2-dimensional case.

- (i) Compute $d_{\text{hyp}}(y_1, y_2)$ for $y_1, y_2 \in \mathbb{R}$, and deduce that $\mathbb{H}^1$ admits an isometric embedding into $\mathbb{H}^2$.
- (ii) Show that $\mathbb{H}^1$ is isometric to $\mathbb{R}$ with the usual norm.
- (★) Is $\mathbb{H}^2$ isometric to $\mathbb{R}^2$ with the Euclidean norm?